

Mehul Kaushik

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EDUCATION

Manipal Institute of Technology
Bachelor of Technology, Computer Science

Manipal, India
2023 – 2027

EXPERIENCE

Head of Cybersecurity
ISTE Manipal

2025 – 2026
Manipal, India

- Coordinated CTF competitions for over 150 participants, crafting challenges on web security.
- Directed a 10-member team conducting workshops and mentorship sessions.

Project Intern
STMicronics

May 2025 – July 2025
Noida, India

- Developed automated reporting workflows using Python, reducing manual effort by 60%.
- Developed modular Python automation scripts for the CVIP PLL verification layer, reducing manual verification effort by 40% and accelerating regression workflows by 2 times.

PROJECTS

MoonMovement – Startup Community Platform | *PostgreSQL, Prisma, Node.js, Redis, AWS* GitHub | Live

- Built a Reddit-style startup community platform supporting startup discovery, threaded discussions, voting, user profiles, startup directories, and role-based moderation through a scalable REST API architecture.
- Designed a 3-tier backend architecture with stateless API servers, PostgreSQL, and Redis caching, deploying the application on AWS EC2 with Nginx reverse proxy and AWS RDS for managed database hosting.
- Engineered low-latency content delivery using Redis for session management, hot-feed caching, and rate limiting while optimizing complex Prisma queries to eliminate N+1 lookups and reduce database load.
- Implemented secure authentication, RBAC, and asynchronous notification workflows, enabling scalable concurrent request handling with efficient database indexing, pagination, and transactional consistency.

RAG-Powered Document Q&A System | *Node.js, MongoDB, Redis, Vector Search*

GitHub

- Built a RAG system that enables users to upload PDFs and query them via a conversational LLM interface.
- Processed documents through a 5-stage async pipeline (parse → chunk → embed → index → search), efficiently handling PDFs up to 20MB and generating 768-dimensional embeddings for semantic vector search.
- Offloaded PDF processing to a decoupled Redis worker using a BRPOP blocking queue, enabling the API to return 202 Accepted immediately while tracking asynchronous job status via Redis Hashes (HSET)
- Engineered hallucination mitigation via similarity thresholding (≥ 0.7 cutoff) to reject low-relevance context before LLM generation, and implemented automated state cleanup to prevent orphan records in MongoDB during async worker failures.

TECHNICAL SKILLS

Languages: C++, Python, Java, SQL

Backend: Node.js, Express.js, REST APIs, PostgreSQL, MongoDB, Prisma, Redis

Cloud & DevOps: AWS (EC2, RDS), Docker, Nginx, Git, Linux

Machine Learning & AI: RAG, LLM Applications, NLP, Vector Search, CNN

ACHIEVEMENTS

- Solved 600+ LeetCode problems and achieved the Guardian badge.
- Secured AIR 318 nationwide in the Amazon Machine Learning Challenge.
- Won 3 Hackathons and was selected as a Smart India Hackathon (SIH) Finalist.
- Contributed to the HumanAI project during Google Summer of Code, with 1 pull request merged.

CERTIFICATIONS

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| Andrew Ng's Machine Learning Specialization | Credential |
| Google AI Professional Certificate | Credential |
| Google Data Analytics Professional Certificate | Credential |
| AWS Cloud Practitioner Essentials | Credential |